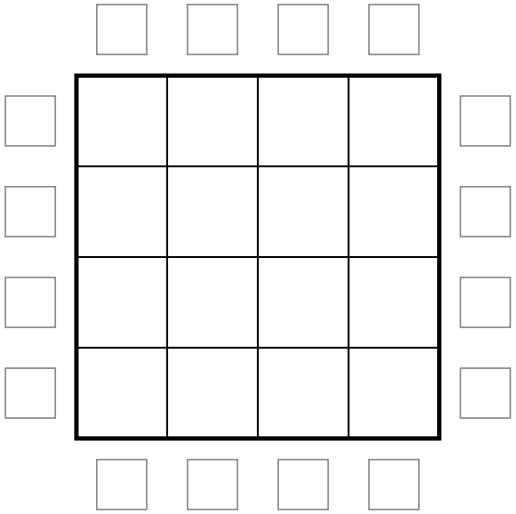
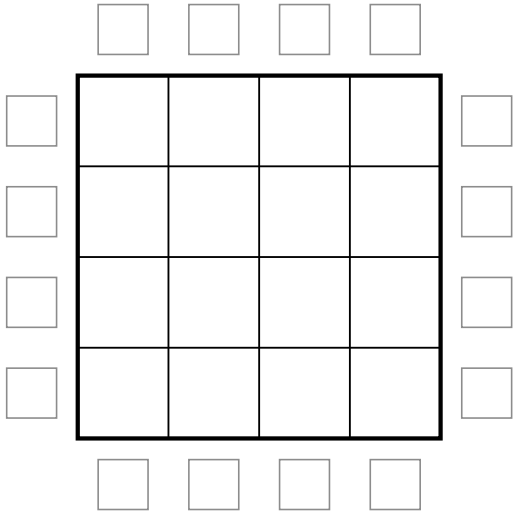
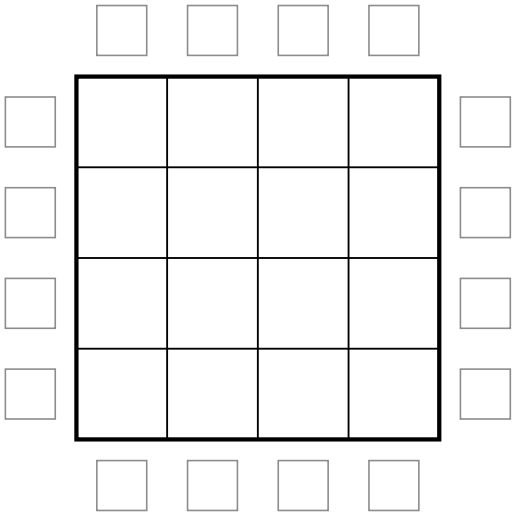
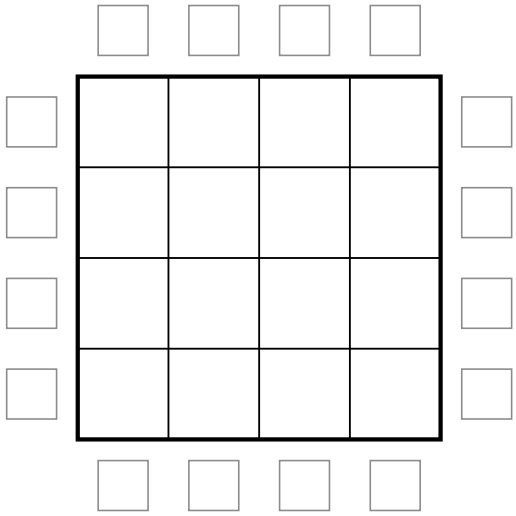
Problem 5: Build a city behind a folder, out of your partner's sight. Write some of its numbers around an empty grid so that your city is the only one that fits, using as few numbers as you can. Your partner solves the puzzle and then tries to find a second city that fits the same numbers.

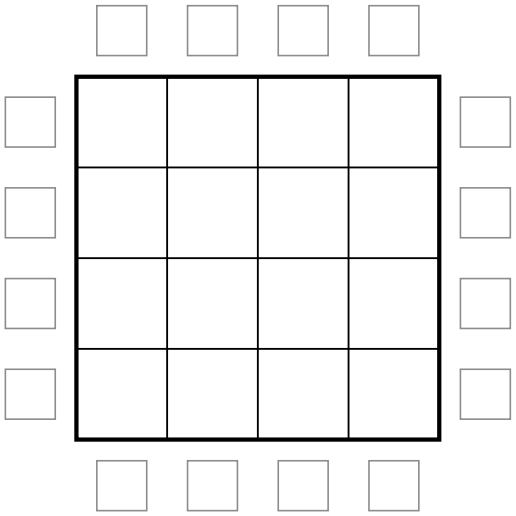


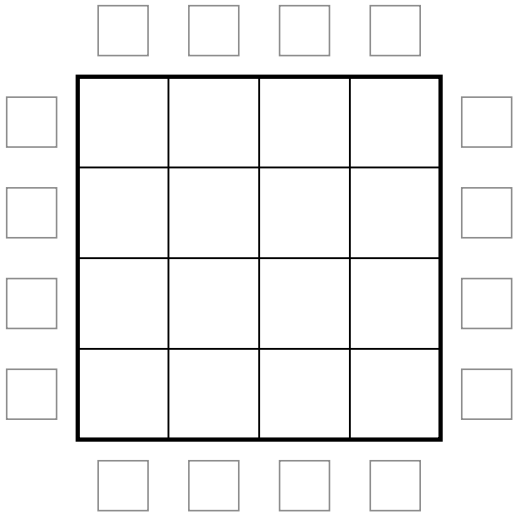






Problem 6: Make a puzzle with only three numbers around the grid, none of them a 4, that has exactly one city.





Problem 7: There are 24 different rows of four towers 1, 2, 3, and 4 cubes tall. Sort them by the two numbers seen from the ends, and write in the table how many rows show each pair.

seen from the left end \ seen from the right end	1	2	3	4
1				
2				
3				
4				

Problem 8: There are 120 different rows of five towers 1, 2, 3, 4, and 5 cubes tall. How many of them show exactly 1 tower from the left end? How many show exactly 2? Explain how you counted.

Problem 9: Can two different 4-by-4 cities have exactly the same sixteen numbers around them? If they can, find two. If not, explain why.

